

互動設計期中報告：記帳軟體調查

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甲、前言

還身為大學生的我們都經歷過那段必須節省花用，以度過月底窮困潦倒的日子。為了避免帳戶不知不覺地沒錢，許多我們周遭的人都選擇使用記帳軟體去了解金錢的流向，但卻都又不約而同地放棄使用記帳軟體。我們希望在期末時製作一個app雛形，藉由遊戲化抑或是產生歸屬感的方式，重啟使用者記帳的初衷。所以在此之前我們需要了解這些人是為什麼不繼續使用記帳軟體？我們應該用什麼方式幫助使用者維持習慣？

乙、測試系統簡介

1. Andromoney

這是我們測試的五款app中相對商業化的一款。它的介面採用接近Android原生的風格，在紀錄花費的過程中能夠選擇使用的幣種、帳戶，並且具備非常多元的帳目分類，適合商務人士到不同國家切換使用。累計圖表分為折線圖、圓餅圖、長條圖等，提供使用者用自身直觀的圖表去了解花費習慣。

2. 碎碎念記帳

這是一款以父母管教督促為出發點而設計出的類遊戲化記帳軟體。它模仿聊天軟體的介面，並讓使用者以傳送對話的方式輸入花費項目；系統接收到資訊後會讓AI模擬的父親回傳訊息，提醒或是評論使用者的消費習慣，進而警惕使用者。此軟體因為融合不同類型軟體間的功能，使得介面複雜沒有邏輯的缺點更加明顯。

3. Ahorro

這款軟體主打介面簡名卻功能完整，曾經是記帳軟體下載排行榜中的冠軍。它以紅色且較為卡通化的主題風格設計，字體較大讓使用者能更快地捕捉重點，並提供圓餅圖分析各種花費種類間的比例。整體上算是五臟俱全、中規中矩的記帳軟體。

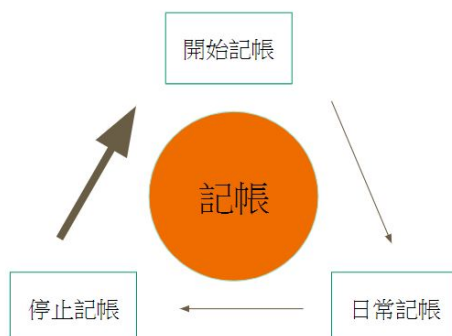
4. 記帳城市

這是一款相較其他記帳軟體不同，帶入遊戲化成分較高的記帳軟體，比如說：輸入現實生活中的花費項目就可以在遊戲中蓋一棟房子，並經營屬於自己的城市。它透過以上的規則等，去和使用者建立約束感，維繫使用者記帳的習慣。它也設計一套主題風格明顯的介面，讓使用者體驗的過程中有整體感，並且提供圓餅圖和長條圖等了解自己的消費比例。

5. Coinkeeper

這也是一款比較有特色的記帳軟體，利用操作手勢去隱喻金錢的流動，讓使用者地記帳過程變得更加生動。它的設計理念是將金錢拖出帳戶以進行記帳，並提供各個支出和投入項目，和整體支出的金額長條圖表。

丙、測試目的



(此為我們這次測驗的概念圖)

主要概念:

了解使用者為什麼開始使用記帳軟體，並探究後來停止記帳習慣的原因是否與記帳軟體介面及設計有關。

主要受試者:

五位使用不同記帳軟體但後來停止記帳的使用者

概念細節:

- 開始記帳: 我們了解受試者為何開始有記帳習慣，是否與記帳軟體有關
- 日常記帳: 開始深入探討軟體設計、使用介面、操作流程等等是否會影響受試者記帳習慣
- 停止記帳: 為什麼停止記帳呢？是否與軟體有關

期望成果:

分析不同受試者是否因不同的記帳軟體而導致停止記帳的習慣，找出大眾的原因，作為我們設計系統介面、操作流程、核心架構的新元素，讓人們可以透過我們的記帳軟體，重拾記帳這習慣。

****為什麼選擇測試五個呢？**

透過五種設計理念、使用者介面、記帳操作流程截然不同的記帳軟體來做比較，分析之間的差異，了解受試者的量化的反應及評比，找出最合適大眾的方案。分析的部分會比較立體的呈現出這裡描述的項目。

丁、測試過程

每位受試者都會經過三個測試階段，分別為(一)介面問卷調查 (二)Think Aloud 測試過程 (三)使用者經驗訪談，讓我們能夠全盤了解受試者記帳的習慣及其使用軟體的過程及感想，以下也會更進一步說明各階段的目的及細節。

測試時間: 10/27-10/29

測試地點: 清華大學校本部 電資館842

測試形式: 每次受試者為一位，再經由我們的測試說明，並同意接受側錄及訪談之後，依序進行(一)-(二)-(三)的測試階段

測試細節說明:**(一)介面問卷調查**

目的:

1. 了解各記帳軟體的介面影響使用者的因素比例。
2. 在使用者進行think aloud測試過程之前，簡單了解使用者在使用軟體的習慣內容(依據目的，將題目設計成兩種，製作成簡單的量化表單):

問卷問題	
我覺得介面很直觀 Likert Scale	使用此軟體多久
我喜歡軟體介面視覺上的設計 Likert Scale	平均上，再用記帳軟體的時候買完東西換延遲多久才會記帳

介面的排版會造成我使用困難Likert Scale	平均上，一天會打開幾次記帳軟體？
我覺得介面很好看Likert Scale	用這個記帳軟體可以幫助我記住我以前做過的事情Likert Scale
我覺得記帳流程很麻煩Likert Scale	記帳軟體讓我意識到我原本不知道跟自己有關的事情Likert Scale
基於介面，我想要繼續使用這軟體Likert Scale	記帳軟體的分類太瑣碎Likert Scale

(二)Think Aloud 測試過程

目的：透過使用者一邊使用一邊講出自身的使用過程去觀察其停頓的原因並分析每個軟體之間的差異。

內容：

1. 實驗操作員先以手機版 Google Map 操作任務「使用導航設定從台南火車站到國立清華大學」並同時進行Think Aloud，讓受試者了解Think Aloud時所應該講出來的話的程度。
2. 告知受試者任務「昨天買了50元的鹹酥雞宵夜，今天買了35元的綠茶。請把這兩個項目按照日期紀錄起來。紀錄玩之後，開啟統計圖表觀看消費紀錄。」
3. 讓受試者以曾經使用過的記帳軟體操作任務，並同時進行Think Aloud。
4. 實驗操作員同時在一旁側錄其使用過程，並給予適時提醒。

(三)使用者經驗訪談

目的：了解受試者的使用經驗及停止記帳的原因

訪談形式：使用手機經由受試者同意後開始錄音並進訪問，過程中使用一問一答的方式，在聽到受試者回答之後，有效地拋問題給受試者，幫助達到測試目的，以不打斷受試者回答為原則，全程記錄受試者的訪談內容

訪談問題：

1. 當初是因為哪些原因想要做記帳這件事情？
2. 當初是因為哪些原因決定用這個軟體？
3. 因為哪些原因停止使用此軟體？
→ 從記帳的開始到停止習慣的背後原因探討
4. 使用這個軟體有影響過你的消費習慣嗎？
5. 軟體裡面的圖表哪些方面對你有用？
6. 哪一類型的圖表讓你最容易了解你的花費狀況？
→ 了解記帳是否真的有用及如何快速達到記帳效果
7. 當時在花費紀錄裡面有看到吸引到你注意力的事情的時候（例如錢花得多或少，花在什麼類型的東西上）有什麼樣子的感覺？=>了解記帳的當下是否會有特別的情緒反應 → 注重在繼續使用是否也能幫助他們記住他們當下的感受，此題為

我們的期末鋪路

8. 你平常有沒有養成其他規律性／持續性的自我提升或娛樂自己的習慣？ 例如：
寫日記、打坐、健身、每日登入獎勵
→ 了解受試者是否有類似記帳的每日習慣，做為參考依據

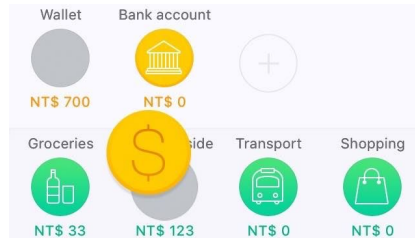
戊、Think aloud測試與問卷 - 分析結果

在這次的實驗中，我們主要紀錄了三種資料包括受測者think aloud操作影片、後側問卷、和訪談。接下來會對每一種資料做分析，最後則是觀察三種資料有甚麼共同點或是有何不同。

1. 受測者操作影片

這次的實驗，每個受測者都會留下影片紀錄，並且在實驗之後，我們會用一台手機的錄影機來記錄受測者所有的步驟，並透過細查影片去觀察每一個操作的步驟——為了發現各種APP有甚麼特色又會造成甚麼問題，以下會按照測試的順序，逐個介紹我們在每個APP的發現。

第一個受測的是，**Coinkeeper**，受測者在完成任務的當中總共使用了**21個操作步驟**，還有**10個錯誤**的次數。這裡將使用者操作的步驟和錯誤分開，是因為當使用者發生錯誤後，通常會需要和錯誤一樣或是更多數目的步驟來修復錯誤，但我們想要觀察不同APP之間的差異。Coinkeeper本身的步驟並沒有特別之處，但是特別的地方是如何開始動作，推需要使用者「拖曳」對應的硬幣到對應的支出項目，其他的APP則是幾乎全部使用「點擊」來完成所有動作。（以下顯示「拖曳」錢包裡的硬幣到其中一個支出項目）



「拖曳」這個動作，比起「點擊」，透過直接視覺化花錢的這個動作，更貼近我們在真正付錢的動作。但是，因為「記帳」和「付錢」是不一樣的動作，所以讓記帳的過程更像付錢的過程會讓使用者需要花更多時間去學習。

第二個APP是，**碎碎念記帳**，完成任務需要**32個操作步驟**來完成任務，並且因為找不到介面上可以改日期的affordance放棄記錄昨天買了甚麼，把兩筆花費都記錄在今天，而且一共出現**31次的錯誤**，大部分都是在尋找如何開啟統計資料，還有另一個原因是嘗試改變日期，以及在尋找統計功能時1次因為發現隱藏的項目分類功能而分心。在完成任務需要的32個步驟中，有2個是在閱讀APP顯示的回話，因為這是這款APP的主要特色，所以也記錄在內。這個APP主要的問題在，設計者希望把所有功能全部都擠在主頁面上，但是因為功能沒有被完善的分類，造成項目太多，而用來提醒使用者的，只有數個小點(有顏色表示目前顯示的功能表，灰色則是其他功能，類似網頁上可以滑動換頁的廣告欄)，如果使用者不仔細看，很可能沒辦法發現這些被隱藏的功能表，即使這些功能都很常見，不管是花費的分類或是統計資料，都需要滑動功能表，切換到別頁，才能使用。

第三個，是現在Google Play上名列前茅的，**記帳城市**。這個APP只需要**17個步驟**來完成任務。受測者發生了一次錯誤，因為實驗用的手機的設定不同，切換鍵盤

輸入法有問題，除此之外，並沒有任何的錯誤。這個APP的重心都放在遊戲性上，但是受測者只專注在實驗任務，而沒有使用遊戲性的部分。

第四個APP是，**Ahorro**，只花了**11個步驟**，沒有發生任何錯誤，也是完成任務最快的，完成任務時間只有最慢的APP的五分之一。其中值得提的是切換日期的方式，使用者可以直接像翻日歷一樣，從首頁透過一次「滑動」就能進入昨天或是明天。這個APP也是用了最少步驟的APP：和第二名的，記帳城市，相比少了6個步驟。

最後受測的是，**理財幫手**，這是一個付費的APP，所以我們只能透過影片來記錄步驟，設測者一共用了**36個步驟**，並發生了**6次錯誤**，都在嘗試尋找統計資料。這個APP是步驟最多的APP，但性質和第二多的相差不多。這個APP也是在任務中出現最多選項的，例如：在選擇花費項目上，就有比較多層階層式的選項，選擇食物這個項目後還會出現更多的細項，午餐、晚餐等等，導致受測者需要多點擊幾次來完成選擇項目的動作。

以上的實驗發現兩個比較凸顯的地方：

1. 使用記帳軟體的人不常使用圖表（在訪談內容裡發現的），所以在操作的時候容易忘記怎麼開。
2. 每個軟體的互動性質並不一定，但可能影響使用者的操作時間。

在5種不同的記帳軟體中，有三個受測者在嘗試開啟統計資料時發生錯誤，而只有Ahorro和記帳城市則完全沒有。可能是因為這兩個APP將功能在主頁做了很清楚能容易用看圖示區分的分類，並且在主畫面上留下按鈕，其他的APP則需要開啟額外的選項、滑動功能業等等步驟才能做到一樣的事。

雖然使用頻率上，我們透過訪談發現圖表似乎不是常用的功能，但似乎都是大部分人認為會存在的功能。碎碎念記帳的受測者，放棄了改變日期的任務，但是沒有放棄開啟統計圖表，可能就是因為認為這個APP應該有這個功能。其他的受試者也一樣，當他們找不到這個功能時，他們表現出疑惑，但是仍然會嘗試尋找，而不會認為這個功能不存在。

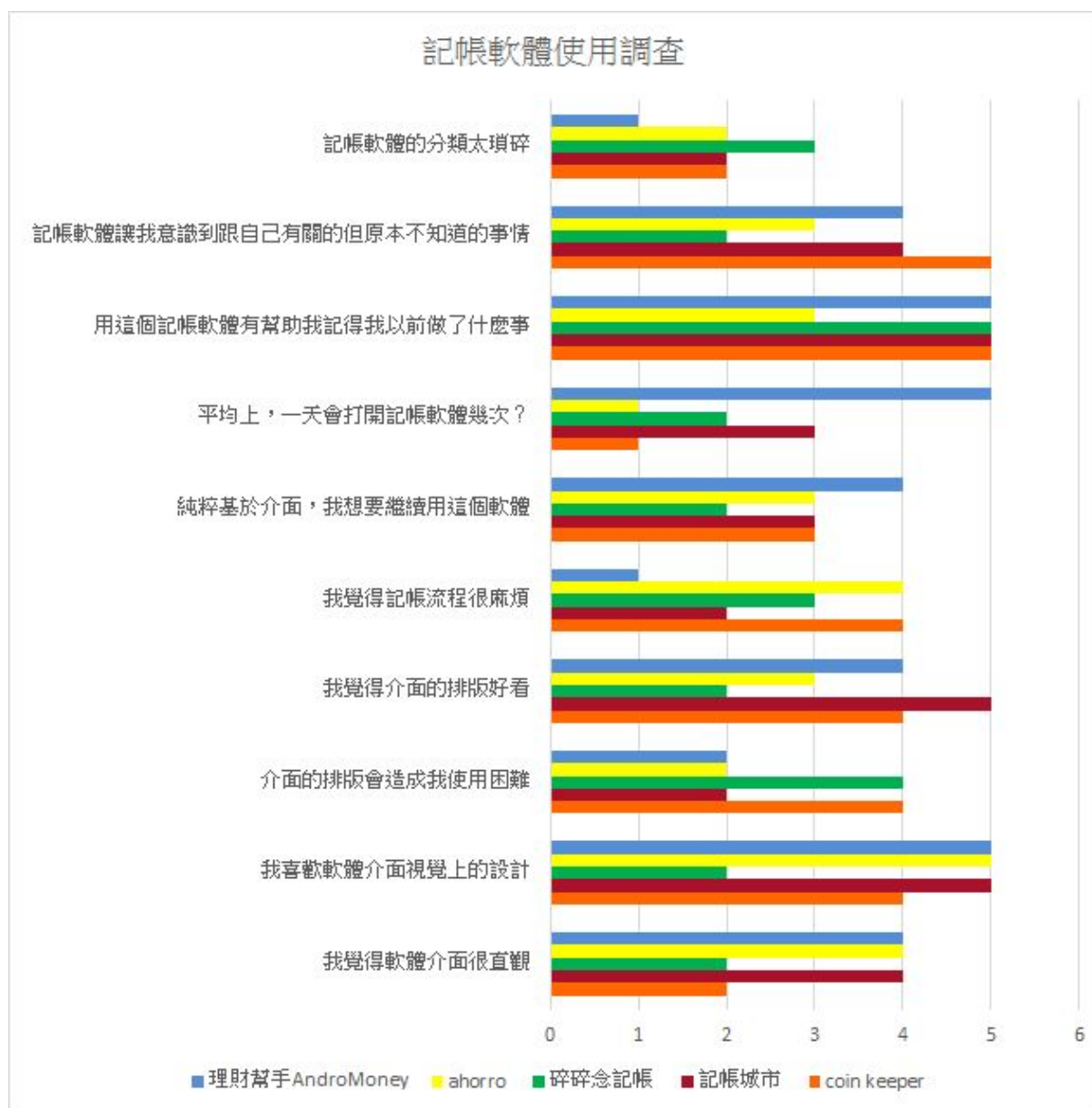
另一個發現是比較記帳城市和碎碎念記帳所發現的問題，碎碎念記帳的互動式送出帳目後的即時回饋，但是這個回饋會讓使用的步驟增加一點（2步驟），當使用者在記帳的同時，也可能會先分心去閱讀電腦的回應再查統計資料。而記帳城市互動性有特色的部分，也就是遊戲化的部分，並不會影響到記帳的過程，使用者可以在完整花費輸入之後再進行遊戲，而不需要中斷動作去回應APP的互動。

2. 後側問卷

使用者測試結束後，我們會讓受測者填寫後側問卷，主要希望能做到的是了解受測者對APP的評價，並希望能符合使用者測試中出現的問題。除此之外，也希望了解使用者記帳的頻率、消費和紀錄的延遲等，因為這些資料沒辦法在這次使用者 think aloud測試中實驗。以下是五種不同APP所得到的問卷結果，以及條形圖表。

	Coinkeeper	記帳城市	碎碎念記帳	Ahorro	理財幫手
1.我覺得軟體介面很直觀（1～5評分）	2	4	2	4	4
2.我喜歡軟體介面視覺上的設計（1～5評分）	4	5	2	5	5
3.介面的排版會造成我使用困難（1～5評分）	4	2	4	2	4
4.我覺得介面的排版好看（1～5評分）	4	5	2	3	4

5.我覺得記帳流程很麻煩（1～5評分）	4	2	3	4	1
6.純粹基於介面，我想要繼續用這個軟體（1～5評分）	3	3	2	3	4
7.以上的軟體用了多久？(月)	24	3	1	30	12
8.平均上，在用記帳軟體的時候買完東西會延遲多久才做紀錄？(hr)	12	4	12	30	3
9.平均上，一天會打開記帳軟體幾次？	1	3	2	1	5
10.用這個記帳軟體有幫助我記得我以前做了什麼事（1～5評分）	5	5	5	3	5
11.記帳軟體讓我意識到跟自己有關的但原本不知道的事情（1～5評分）	5	4	2	3	4
12.記帳軟體的分類太瑣碎（1～5評分）	2	2	3	2	1



後側問卷的資料和使用者think aloud測試所看到的大多相吻合。首先是Coinkeeper，因為使用了「拖曳」的動作來開始記帳的行為，所以在第一個問題上，只拿到兩分。

接下來是碎碎念記帳，我們發現受測者沒辦法有效地找到需要的功能，因為有太多功能被隱藏在主畫面裡，所以在第一個問題也只拿到兩分。除此之外我們也發現，在第9個問題，延遲記帳的情況中，相較於記帳城市(兩者都為記帳這個行為提供額外的動機，碎碎念記帳提供聊天的回饋，記帳城市則提供遊戲性的回饋)並沒有辦法讓使用者更常打開APP。

記帳城市，最能刺激使用者經常打開APP，因為從訪談裡發現這個APP遊戲性的回饋是有時效性的，所以使用者較常會去做記帳這個動作（一天平均三次）。我們想要在意打開APP的頻率的一個原因也是因為導致使用者想多打開APP跟他們養成在花費的當下記帳的習慣有關係，也會導致使用者跟記帳的APP有比較多緊密的關係，讓他們繼續使用的機率升高。

第四個APP是，Ahorro，在關於介面的問卷問題上都得到高分，也證明了Ahorro的簡潔設計和現在人所想要的介面屬性十分吻合，但是也是在花費完後延遲最長才做紀錄的一個軟體。不過它也是相對其他四個軟體被使用過最久的APP。雖然沒有額外的刺激，但是簡單的紀錄方式似乎才是讓使用者持續記帳的最佳方式。

最後是，理財幫手的階層式分類選項，在第12題「記帳軟體的分離太瑣碎」中是在這項目拿到最低分的APP，也代表理財幫手的使用者覺得此軟體的分類最不瑣碎。我們在理財幫手的think aloud分析裡面也有提到因為這個軟體的花費項目分類方式是階層式的而且分的比較細，紀錄花費的時候可能會需要（相對其他記帳軟體）比較多點擊的步驟。不過理財幫手的受試者有在被訪談的時候提到他反而覺得理財幫手多階層的分類方式可以讓他更有效率的分類花費項目，因為此軟體選擇的詳細分類還有它把這些分類區分成階層的方式在受試者的眼裡是又詳細又適合的。因為使用者在其他記帳軟體可能沒辦法將所有帳目放到自己覺得適合的類別，但是每次都要新增類別又會使類別變得太瑣碎，像是有一些記帳軟體只有一個「食物」的分類，但是理財幫手有在「食物」地下自己提供「宵夜」的項目。因此，階層式項目分類似乎是一個不錯的解決辦法，讓使用者可以對項目做更詳細但有一致性的分類。

關於理財幫手的另一個有趣的發現，就是雖然我們可以用記帳城市的時效性遊戲機制去解釋為什麼使用者會一天打開軟體比較多次（3次，比起其他軟體1-2次），我們沒辦法用相似的原因去解釋為什麼理財幫手使用者一天打開軟體的次數多其他軟體蠻多的（5次）。不過我們在訪談裡面有發現理財幫手的使用者原本就有維持規律性在買東西的當下做紀錄的習慣。所以我們也能從這點知道不只能考慮記帳軟體本質上的屬性，但要考慮記帳軟體屬性和使用者原本個性的互相作用(interaction)。

我們基本上也能從圖表裡面綠色的條形看出碎碎念記帳在介面上相對所有測試的其他軟體在介面使用性項目上都是得到最不佳的評分。這也跟我們think aloud測試的發現一致，因為碎碎念記帳是操作步驟最多（32次）而且它的錯誤跟操作步驟幾乎是一樣多的（31次）。

從問卷結果我們也可以看出一些其他值得關注的發現。我們光看圖表就能看出第十題「用這個記帳軟體有幫助我記得我以前做了什麼事」的平均分數是每一個問題最高的，幾乎每一個受試者都給了滿分而且唯一沒有給滿分的也至少答了三分。這其實是一件蠻重要的事情，因為甚至我們測試的受試者都是使用不同記帳軟體的，幾乎大家都有覺得用記帳軟體可以幫助他們記得以前做的事。因為我們的期末主題有接觸到增加使用者和軟體內容的連結感（例如，讓使用者特別注意他們歷史上的某些習慣的存在能促進self awareness跟提升behavior change的表現，導致受試者的精神健康

被提升)，從問卷結果知道了很多種記帳軟體都有幫助到受試者特別記得他們以前做的事情，我們可以把記帳軟體當作一個「記憶提取機制」來做更深入的功能開發。

戊、Interview Findings

At the end of the experiment, we carried out an interview with the participant, in which we guided their response with the questions mentioned in the experiment procedure (測試過程) section of this report. We recorded and manually transcribed the audio from the interview, and categorized the transcribed responses by question and application type for better comparative analysis.

In our analysis, we found that all of the participants listed the curiosity and desire to better understand themselves as part or all of the reason why they wanted to start doing accounting in the first place. This is similar to the motivation for why many people like to do personality tests online - studies speculate that the motivation stems from a human desire to stabilize the characteristic of oneself as part of one's identity, and in turn become closer to oneself. But the reason that some participants, in particular male ones, stop using the app is also directly linked the reason why they started. Both of our two male participants expressed that even after using the accounting app for a while, they felt that they already understood their own spending habits well regardless of whether they kept track of their spending or not, so they felt that the accounting app did not help them understand themselves any better than they already did and stopped accounting for this reason.

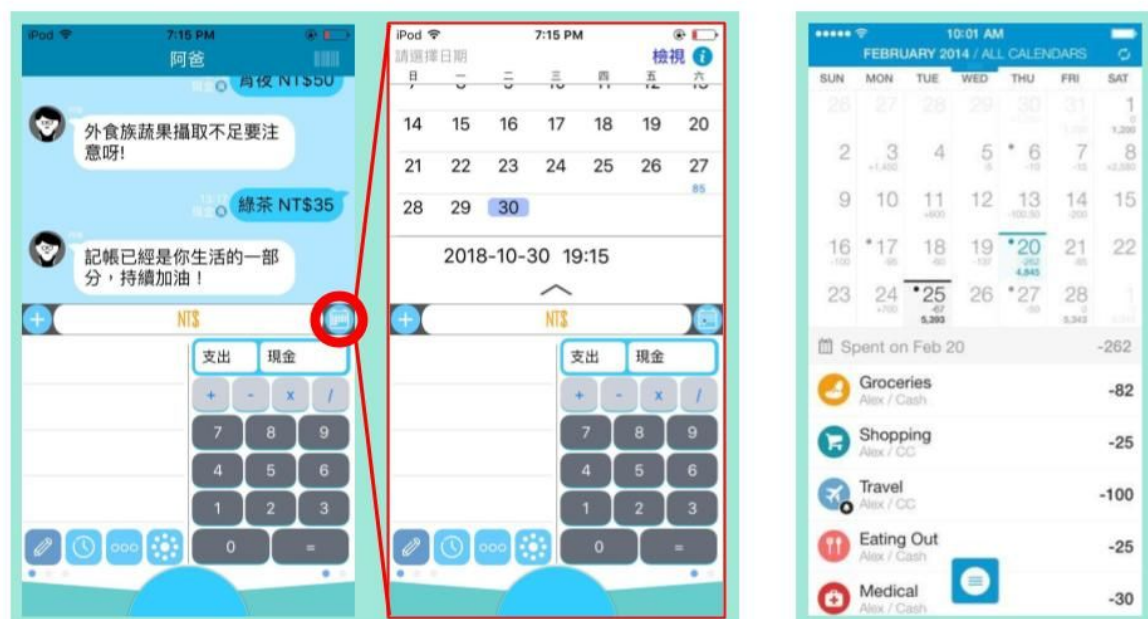
But there were other varying motivations for using accounting apps in the first place. The user of Coinkeeper expressed that he had no internal motivation to use the app, but it was rather because he had started working at a job and his family restricted his spending, so he started accounting to manage his spending habits. This was more of an external motivation. Again here, the motivating reason was directly linked to the stopping reason - he stopped accounting when he quit his job and went back to school, since at that point his spending was no longer restricted and the initial motivation was gone. Some other internal motivations apart from the pure curiosity to understand spending, was mainly the desire to explicitly restrict spending for two reasons: actual lack of money, and desire to simply maintain good and frugal general spending habits.

An interesting finding from the interviews was regarding the users' perceived importance of statistic visualization of the spending history (which occurs mainly in the form of bar graphs, line graphs, and pie charts). In the interview we asked some specific questions pertaining to whether and how users felt that the statistic visualizations in the apps helped them, as well as what kind of visualization graph they found to be most useful. The response that was interesting was a mostly unanimous 「我不常用圖表」 from the users when asked if they found the in-app graphs helpful. But this conflicts with what we see in other results. Precisely, we know that all participants use graphs, because it is the only way they can view their spending history - and we know that they do utilize knowledge of their spending history from the relatively high-scores in questions 10 and 11 from the questionnaire (用這個記帳軟體有幫助我記得我以前做了什麼事 and 記帳軟體讓我意識到跟自己有關的但原本不知道的事情). We even have users say things in the interview such as “「看到花費

紀錄」是可以幫助我反思不見的錢到哪裡的方法”，which was said by the 碎碎念記帳 participant.

Since the graph visualization of spending is the only form of spending history available in the apps, we know that in order for users to say that they feel the accounting app helps them to remember what they've done in the past and realize aspects about themselves they had not previously known, they must have looked at the visualization at some point found that it helps them remember things and learn about themselves, regardless of whether this occurred consciously or not. We also found from the response to the interview question (哪一種類型的圖表讓你最容易了解你的花費狀況?) that users do utilize the statistical visualization. Across all 5 apps, all 5 users responded to this question saying the pie graph (圓餅圖) was best because they can easily see the ratio of their spending over the course of a month-long time period. All users expressed that they care about their general spending ratio (whether they spend more on some things than others) compared to change over time of spending. The above information shows us that app visualization of spending history is meaningful for users, but that they do not explicitly recognize its significance.

We also obtained some useful personal insights from the users about what they thought of spending history visualization. The 理財幫手 user made a point that 理財幫手 includes a type of spending history visualization that most other accounting apps actually don't: a monthly *calendar* that shows your spending and balance on a day-by-day basis, whilst visually highlighting days or times when the spending may have been a lot (see the **right section** of the figure below for a screenshot of this visualization). This is can be a great benefit to accounting app users, because it is a way to visualize *spending over time*, or temporally-focused information in a way that is more fitting to user's **mental model** than a line graph (折線圖), which is usually used to display spending change over time.



Almost all users expressed in the interview that they didn't use or like line graphs for a few reasons. Some users said that looking at the line graph didn't give them any particular feeling about their spending over time, because looking at the

line over and over again makes them feel like the temporal spending patterns are the same. However, the user of 理財幫手 who introduced this calendar feature stated clearly why the calendar worked better for her to display time-based information than a line graph. She said that to glean useful information from a line graph displaying change, she has to look for notable fluctuations in what really just looks like a line, and then match that line fluctuation to what she might have done to, say, spend more money during the time point of the fluctuation. This requires many levels of mental abstraction to get to the point of knowing what you might have done on what day(s) to spend so much money.

But with directly viewing a calendar, humans are familiar with the **mental model** of a visual calendar and so can easily at a glance see what days of the month and what days of the week they spend more money on. Additionally, once they see the exact *date* they can much more easily reconstruct the events of that day than through trying to interpret changes in a line graph. The temporal spending data is much more accessible through a calendar format and more effectively conveys the level of specificity in information needed by the user as well as chronological context.

However, if we look at the left section of the above figure, we can see that the quality of execution in app development is just as important as including the effective concept of a calendar visualization. On the left section is actually the same kind of spending calendar in 理財幫手, but it is very poorly executed and difficult to find. This feature is included in the 碎碎念記帳 app, but like we mentioned earlier the app was difficult to use in many respects and even this feature was something we only found after digging into the interface for a very long time. The user of 碎碎念記帳 who used the app for months did not even know this feature existed. This example just shows us that the affordance of an interface to enable access to important features is just as important as the intention to include the feature itself.

The interview also allowed us to find that both “external motivation”-focused apps, 碎碎念記帳 and 記帳城市, lost their users for similar reasons. We define “external motivation” here as providing motivation to use the accounting app that is separate from accounting itself. In 碎碎念記帳, the user expresses that the app provides immediate socializing feedback (立即回饋) to the user in the form of some very intimate and familiar messages from a chatbot who is supposed to be like a family member (碎碎念爸爸／媽媽). This familiarity actually motivates the user to use the app for a while, but after the messages start repeating the user loses the feeling of social rapport with the application and becomes bored, which leads her to stop using the app. 記帳城市 in comparison is simply a city-building game that allows you to level up the city with consistent accounting. But the mechanics (機制) of the game are very basic and playing the actions of the game becomes boring after a while, which also causes the user to lose interest and stop using.

Lastly, we also unveiled some interesting potential gender differences in the interview feedback. But because the sample size was very small, we want to emphasize that these just happen to be some trends that we found to prevail in our user sample and likely may not represent the general population. In addition, everything we found in the interview comes as self-reported information and has more to do with how the users’ perception of themselves than any tangible behavior.

Our study involved 2 male users and 3 female users. Both male users expressed that even after using the accounting app, they felt that all of their spending amount, patterns, and habits were within their expectations - they expressed that everything they did was to their expectation and they didn't need to change much. But all 3 female users self-reported instances or habits in which they would *realize* that they had spent too much money on something, or had been spending more money than they expected. Then they would express that after the shock they would want to cut down on spending. None of the 2 male users expressed surprise, even though they mentioned in the interview that there were still one or two times where they would "notice changes and look into recent spending".

The female contrast in how realization of spending too much money is expressed can be seen in all three female users' speech. The user of 碎碎念記帳 said, "我領完錢 但發現很快就沒有了 而且不只為什麼 感覺錢被偷了但其實沒有人偷的感覺," which expresses a form of surprise in feeling as if the money was stolen. The user of 理財幫手 said, "大學的時候很常跟朋友去吃飯 在食物的類別發現花了特別多錢 然後會嚇到," using the phrase 「嚇到」 to express her shock. The user of 記帳城市 said, "會發現原來在這裡花這麼多錢 有點驚訝," also expressing surprise. In contrast, male users tended to talk about their spending changes like "如果發現花費有變化, 會回想那筆錢花在什麼東西上。如果沒什麼特別的, 就不會去理它," or "若本來就是打算砸一筆錢在甚麼東西上 那這種東西我就是不太Care," both expressing more lack of feeling about their spending in general.

Males also had higher self-perceived ability of controlling their spending. Both expressed similar thoughts along the lines of, "後來覺得有紀錄還是會花費一樣的價錢, 開始覺得記帳沒有甚麼用, 自己本來就會控制花費。" These findings may just be due to sampling bias, but if not we are unsure whether the difference results more from actual behavior or self-perception.

己、Usability Problems

We will split the summary of usability problems into interface-focused problems and non-interface problems. Interface-focused problems are issues with purely the interface of the application that may make it difficult to use. Non-interface problems are more related to overarching issues of user-app interaction covered in questionnaire and interview analysis.

The main interface problems we found were with 碎碎念記帳 and Coinkeeper. In 碎碎念記帳, as can be seen from the figure on page 9, compression of the screen space due to the messaging box taking up half of the whole screen makes the space available for making app functionality visible very small. Much of the features are hidden in a small rectangular area containing button-like icons near the bottom left of the screen, which you also have to swipe through in order to access all available functions. This violates the usual app design rule of making all important functions directly available on the home screen.

In Coinkeeper, there are very serious mistakes such as misleading icons and menu labels that are not understandable. The misleading icon occurs in the spending history visualization: there is an icon for each different kind of graph that you want to show. When you tap on the pie graph icon, it shows a pie graph. But when you tap on the line graph icon, it actually shows a bar graph. There is actually no line graph

available in the app, which confuses users by showing a representation of something it doesn't actually do. The statistic visualization is also extremely hard to find in Coinkeeper, as it is hidden inside a menu option called "**Feed**". Neither the user themselves or other experimenters trying the app were able to know what it meant before opening it to reveal a hodge-podge of statistical information types in no particular organization. This not only fails as an affordance to the user, but also displays information in an incomprehensible way. Now, we will move onto the non-interface problems.

We have seen in both the external-motivator and "plain version" accounting apps evidence that users become bored due to the repetitive nature of the interaction. In the accounting apps without external motivator (ahorro, Coinkeeper, 理財幫手) the users will stop using the app by themselves if they lose their original personal motivation, as the app provides no motivation. Making spending entries every day is also a repetitive and boring task, so unless there is another reason nobody will want to do this, that is clear. But even for external-motivator apps such as the gamified 記帳城市 and the social rapport-building 碎碎念記帳, the apps themselves are unable to change the mechanics or nature of the interaction even if there is variety across individual instances of using the app - for example, building a hospital instead of a grocery store in 記帳城市 or getting a different message from 碎碎念爸爸. But like the users said, over time, there are limited kinds of buildings you can build in 記帳城市 and a set number of different messages you can receive from 碎碎念爸爸. There is no pure generative nature in the app that allows it to truly change over time, so the inevitable repetition causes users to leave.

Another issue is the perception of spending history visualization brought up in the interview findings section. Users need this visualization of spending history in order to get more meaningful information about themselves, continue to find the app useful, and learn what they need to know in order to conduct any needed spending behavior change. But users don't seem to prioritize this feature, because they only mention that they just don't use it often even though it seems to be an important part of the purpose of accounting.

The "law of inertia" (慣性) phenomenon is also something we observed with the 理財幫手 user. In the interview, she revealed that the reason why she stopped using the app was due to something similar to inertia. During the period of time she was using the app, this user adhered strictly to a routine of opening the app and making a spending entry whenever she spent money, which is why her average "times app opened per day" is 5, much more than any of the other apps. She said that once she creates a routine for herself like this, under normal circumstances she will faithfully adhere to it indefinitely. But if her environmental circumstances change, a large change at a point in time tends to break her habit. After her habit is broken, it is difficult to start the habit back up again.

There are two examples. When she was using 理財幫手, the time she stopped using it was when she was very busy preparing for a conference in Tainan and did not have time in the day to make her usual spending entries. She told herself that after she returned to Hsinchu (about two weeks after she had to stop making entries on time), she would start up again. However, because she stopped that one time during the conference for two weeks, she could not get herself to start up the

habit that she had kept for a year straight again even after going back to Hsinchu. The other example of this “inertia” phenomenon is from our “你平常有沒有養成其他規律性／持續性的自我提升或娛樂自己的習慣？” interview question. When answering this, she mentioned that she used to use a daily exercise app every day for a year. The time when she stopped doing this was when she became very busy with semester finals, and stopped temporarily - she couldn’t start the habit again even after finals ended. So we can see that even for people who may have very faithful accounting habits, it’s possible that the occurrence of a large disruptive event in their life may permanently break a habit they had been keeping.

庚、Revision & Design Insights

We will first address the simple interface problems brought up for 碎碎念記帳 and Coinkeeper. In 碎碎念記帳, we advise not having the chat viewable at all times, also taking up the home display of the app. The home display of the app should prioritize its space in facilitating the initial steps of using the app, which don’t include viewing the chat. The chat is viewed each time after a spending entry is made, so we would advise maximizing all screen space to clearly show all of the basic functions of the app (e.g. create spending entry, view spending history, settings, view calendar, etc.) clearly on the home page. Since “create spending entry” is an option on the home page, we can pull up the chat view (that is in the page 9 figure) only when the user actually decides to create an entry. This way the user can still use the app with its intended purpose of filling in entries as if sending text messages and getting an immediate response from the chatbot.

The Coinkeeper issues are much more simpler to solve. The misleading icon of a line graph for a bar graph should just be replaced with a bar graph, and the confusing label of “Feed” in the app’s side menu should be replaced with “Spending History” or something of similar meaning. The types of information displayed should also be labeled and also shown in a “tiled” fashion where the user can see all types on the screen at once, instead of its original “scrolling” form where the user has to scroll down to see everything.

Now we will address the non-interface problems in the previous section, as well as suggest some important design insights that came from the interview results. We will first go over ways to fix the repetition-induced boredom that occurs to users even in “external-motivator” accounting apps. Our first approach is to integrate **storyfication** (故事化) into the app. People become engrossed in and want to read stories to the end because there is a continuously evolving plot that changes in an unpredictable way over time. The concept of this is that in a potentially gamified app, storyfication in the mechanics of the game will motivate the user to keep playing because they really don’t know what will happen in the future. In 記帳城市, the city expands in a predictable way when you make more entries and build bigger buildings. But if the further you play into the game, different things start happening to the city based on the type of thing you build, there is more incentive for the user to explore how their input can actually change the game itself. In 碎碎念記帳, the concept of storyfication can be applied more in the sense of implementing interactive machine learning in the chatbot which would make it change its social behavior with the user over time based on the user’s unique inputs, so this would remove the

repetitive nature of the chat messages. Expanding **freedom of choice** for the users of the app is also a way to reduce boredom, such as not limiting the kind of building users can build in 記帳城市 to make it more creativity-based - this would motivate the user to mold the app to their own liking. People's motivation to play games like Minecraft for long periods of time result from the same kind of freedom of choice.

There are two ways we can improve the use spending history visualization, which seems unappreciated. Based on user feedback, more visual and interface priority should be placed on pie graphs because ratios are what users want to see the most, and what they find most useful. Calendar-based temporal visualization of spending history should also be used in place of line graphs, since no user seemed to appreciate the temporal information from line graphs but user mentioned the reason behind effectiveness of calendar-format temporal visualization of spending. Secondly, we should implement periodic reminders to remind users to look at the spending history after suitable time intervals. This is because users said they don't look at the history often, even though they do look at it. Since we know that this is not something users plan to look at often, but that they still need to use, we can, for example, on a monthly or bi-weekly basis send app reminders to remind the user to look at the history. If users know that this function exists, the mental burden of reminding themselves to look at the app is removed and perhaps they can appreciate the information this history provides without feeling like having to check for it is troublesome.

The approach of periodic reminders can also be tested to alleviate the issue of "law of inertia". Because the user was unable to start opening the app again by themselves after a sudden period of being forced to stop using, the internal trigger to start the habit did not work. In this case perhaps an "external trigger" such as app notification can be used to more explicitly trigger the user to open the app again. The user expressed that she wanted to start, but couldn't get herself to do it. So if there is an event similar to "someone telling her to do this thing", she can be aided by the external motivation. Of course, if we use periodic reminders after the accounting app is not used for a long time, we will implement exponential backoff (指數後退) which gradually stops notifying over time in order to avoid annoying the user.

辛、Reflection

The main limitations for our study here were the sample size and duration of the study. Because our topic application is an accounting app, a short user study doesn't actually demonstrate the true usability of the app. Accounting apps are meant to be used over a long period of time, which we were unable to do a think-aloud observational study for with our restrictions. For example, the nature and richness of the information the user accumulates in the accounting app over time differs a lot from what the user gets after a short one-time entry. A way to improve would be to conduct a longitudinal study in which the user would regularly report back regarding some aspects of their usage.

The sample size is clearly an issue for this experiment, with 5 users total and 1 user per each application type. We tried to mitigate this by having them do in the user study the same basic task that could be done on all of the applications. But we feel that despite doing the same user study procedure over five different accounting

applications, we still got some very rich and comparable data which is expressed by the analysis section. We think that if there were more resources, it would be ideal to explore across a larger number of users per application type (e.g. 25 users) more features of each application through custom extensions of the “base task” for more live usability testing.

The think-aloud data was also a bit difficult to analyze because of users’ individual differences. Even after explaining how to do a think-aloud task and demonstrating a live example, the level and type of detail that users expressed in their think-aloud differed significantly. We also did not force or control users to adjust their execution of the think-aloud, but perhaps some small cues could have been made. This made the interpretability of each user’s mental process during the task operation less affordable.